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TREND-BASED MOCK PAPER

GATE Electrical Engineering

Mock Paper — 02

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Electrical Engineering
Paper:	Mock 02
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Choose opposite of 'latent':

- (A) Obvious
- (B) Hidden
- (C) Clear
- (D) Manifest

Q2. [1-NAT] [1 mark]

Speed = Distance /:

(Numerical Answer Type — Enter the numerical value)

Q3. [1-NAT] [1 mark]

Distance 200km at 50km/h takes:

(Numerical Answer Type — Enter the numerical value)

Q4. [1-MCQ] [1 mark]

Choose synonym of 'pragmatic':

- (A) Idealistic
- (B) Practical
- (C) Theoretical
- (D) Abstract

Section B: Electrical Engineering

Q5. [1-MCQ] [1 mark]

Eigenvalues of identity:

- (A) All zero

- (B) All one
- (C) Distinct
- (D) Depends

Q6. [2-MSQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q7. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q8. [2-MCQ] [2 marks]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q9. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q10. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q11. [2-NAT] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q12. [2-MCQ] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{(-\lambda)}(1+z)$

Q13. [2-MCQ] [2 marks]

The vector (1,2,3) projected onto (2,1,0) is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) (2,1,0)
- (D) (1,2,3)

Q14. [1-NAT] [1 mark]

Slip in induction motor:

(Numerical Answer Type — Enter the numerical value)

Q15. [1-MCQ] [1 mark]

Alternator produces:

- (A) DC
- (B) AC
- (C) Both
- (D) None

Q16. [1-MCQ] [1 mark]

Slip $s =$

- (A) $(N_s - N_r)/N_s$
- (B) $(N_r - N_s)/N_s$
- (C) N_s/N_r
- (D) N_r/N_s

Q17. [1-NAT] [1 mark]

Transformer works on:

(Numerical Answer Type — Enter the numerical value)

Q18. [1-MCQ] [1 mark]

DC starter used to:

- (A) Regulate speed
- (B) Limit start current
- (C) Reverse
- (D) Increase torque

Q19. [2-MCQ] [2 marks]

6600/660V transformer turns ratio:

- (A) 10:1
- (B) 1:10
- (C) 100:1
- (D) 1:100

Q20. [2-MCQ] [2 marks]

Over-excited synchronous generator operates at:

- (A) Lagging PF
- (B) Leading PF
- (C) Unity PF
- (D) Zero PF

Q21. [2-MCQ] [2 marks]

DC shunt motor: $V=220V$, $R_{sh}=220\Omega$, $R_a=1\Omega$, $I=50A$. Back emf:

- (A) 171V
- (B) 219V
- (C) 220V
- (D) 225V

Q22. [3-MCQ] [3 marks]

3-phase transformer primary:secondary turns = 10:1. If primary is delta, secondary is:

- (A) Delta
- (B) Star
- (C) Both
- (D) Either

Q23. [1-MCQ] [1 mark]

Per unit system base values:

- (A) Voltage only
- (B) Power only
- (C) Voltage and Power
- (D) Current

Q24. [1-MCQ] [1 mark]

Synchronous generator excitation:

- (A) Rotor
- (B) Stator
- (C) Both
- (D) None

Q25. [1-NAT] [1 mark]

Circuit breaker operates on:

(Numerical Answer Type — Enter the numerical value)

Q26. [1-MCQ] [1 mark]

Most efficient transmission:

- (A) Low voltage
- (B) High voltage
- (C) DC
- (D) AC

Q27. [2-MCQ] [2 marks]

100km line: $R=0.1$, $X=0.5 \Omega/\text{km}$. Short line ABCD:

- (A) $A=1, B=60, C=0, D=1$
- (B) $A=0.98, B=j50, \dots$
- (C) $A=\cosh(gl), \dots$
- (D) $A=1, B=10, C=0, D=1$

Q28. [2-MCQ] [2 marks]

3-phase, 4-pole IM at 1440 rpm on 50 Hz. Slip:

- (A) 2%
- (B) 4%
- (C) 6%

(D) 8%

Q29. [2-NAT] [2 marks]

Transmission line: sending end voltage higher than receiving for:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-NAT] [2 marks]

Gauss-Seidel converges when matrix is:

(Numerical Answer Type — Enter the numerical value)

Q31. [2-MCQ] [2 marks]

6600/660V transformer turns ratio (primary:secondary):

- (A) 10:1
- (B) 1:10
- (C) 100:1
- (D) 1:100

Q32. [1-NAT] [1 mark]

Full-wave rectifier uses:

(Numerical Answer Type — Enter the numerical value)

Q33. [1-MCQ] [1 mark]

Step-down chopper output:

- (A) Less than input
- (B) More than input
- (C) Equal
- (D) Zero

Q34. [1-MCQ] [1 mark]

Thyristor is:

- (A) Current controlled
- (B) Voltage controlled
- (C) Power
- (D) None

Q35. [1-MCQ] [1 mark]

Inverter converts:

- (A) DC to AC
- (B) AC to DC
- (C) DC to DC
- (D) AC to AC

Q36. [2-NAT] [2 marks]

IGBT combines features of:

(Numerical Answer Type — Enter the numerical value)

Q37. [2-MCQ] [2 marks]

CSI input is:

- (A) Voltage source
- (B) Current source

- (C) Both
- (D) Neither

Q38. [2-MSQ] [2 marks]

Full-wave rectifier PIV across each diode:

- (A) V_m
- (B) $2V_m$
- (C) $V_m/2$
- (D) $\sqrt{2}V_m$

Q39. [2-MSQ] [2 marks]

For a buck converter, $V_{out} =$

- (A) DV_{in}
- (B) V_{in}/D
- (C) $V_{in}/(1-D)$
- (D) $V_{in}D/(1-D)$

Q40. [2-NAT] [2 marks]

Step-down chopper $V_s=200V$, $\alpha=0.6$. Output:

(Numerical Answer Type — Enter the numerical value)

Q41. [1-MCQ] [1 mark]

Time constant of RC:

- (A) RC
- (B) R/C
- (C) C/R
- (D) $1/RC$

Q42. [1-MCQ] [1 mark]

Root locus plots:

- (A) Open loop poles
- (B) Closed loop poles
- (C) Both
- (D) None

Q43. [1-MCQ] [1 mark]

Integral controller eliminates:

- (A) Rise time
- (B) Steady state error
- (C) Damping
- (D) Overshoot

Q44. [1-NAT] [1 mark]

Gain margin:

(Numerical Answer Type — Enter the numerical value)

Q45. [2-NAT] [2 marks]

Gain margin is defined at:

(Numerical Answer Type — Enter the numerical value)

Q46. [2-MCQ] [2 marks]

Integral controller eliminates:

- (A) Rise time
- (B) Steady-state error
- (C) Damping
- (D) Overshoot

Q47. [2-NAT] [2 marks]

Phase margin is defined at:

(Numerical Answer Type — Enter the numerical value)

Q48. [2-MCQ] [2 marks]

A system with characteristic $1+s = 0$ has:

- (A) Stable
- (B) Marginally stable
- (C) Unstable
- (D) Cannot determine

Q49. [2-MCQ] [2 marks]

Thevenin equivalent with $V_{th}=10V$, $R_{th}=5\Omega$, load= 5Ω . Output:

- (A) 5V, 1A
- (B) 5V, 0.5A
- (C) 2.5V, 0.5A
- (D) 10V, 1A

Q50. [2-MCQ] [2 marks]

Two-port h-parameter h_{11} has units:

- (A) Ohm
- (B) Mho
- (C) Volt
- (D) Dimensionless

Q51. [2-MCQ] [2 marks]

Superposition theorem applies to:

- (A) Linear only
- (B) Non-linear
- (C) Both
- (D) Time-varying only

Q52. [1-MCQ] [1 mark]

Negative feedback:

- (A) Increases gain
- (B) Decreases gain
- (C) No effect
- (D) Oscillates

Q53. [1-MCQ] [1 mark]

Ideal op-amp gain:

- (A) Infinite
- (B) Zero
- (C) Unity

(D) Negative

Q54. [1-MCQ] [1 mark]

RC phase shift uses:

- (A) 180°
- (B) 360°
- (C) 90°
- (D) 45°

Q55. [1-MCQ] [1 mark]

ICBO in transistor is:

- (A) Collector-Base reverse
- (B) Emitter-Base
- (C) Collector-Emitter
- (D) None

Q56. [1-MCQ] [1 mark]

Aliasing avoided by:

- (A) Increasing rate
- (B) Decreasing rate
- (C) Both
- (D) None

Q57. [1-MCQ] [1 mark]

LTI characterized by:

- (A) Impulse response
- (B) Step response
- (C) Frequency
- (D) All

Q58. [2-NAT] [2 marks]

FS coefficients:

(Numerical Answer Type — Enter the numerical value)

Q59. [2-MCQ] [2 marks]

Displacement current added by:

- (A) Faraday
- (B) Maxwell
- (C) Ampere
- (D) Gauss

Q60. [2-MCQ] [2 marks]

Corona loss increases with:

- (A) Wire diameter
- (B) Voltage
- (C) Both
- (D) None

Q61. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q62. [2-MCQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q63. [1-NAT] [1 mark]

PLD stands for:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-MCQ] [1 mark]

Universal gate:

- (A) AND
- (B) OR
- (C) NAND
- (D) XOR

Q65. [1-MCQ] [1 mark]

JK flip-flop toggles when:

- (A) $J=K=0$
 - (B) $J=K=1$
 - (C) $J=0, K=1$
 - (D) $J=1, K=0$
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	A	1
2	1NAT	A	1
3	1NAT	B	1
4	1MCQ	B	1
5	1MCQ	B	1
6	1NAT	A	1
7	1MCQ	B	1
8	1MCQ	A	1
9	1NAT	B	1
10	1MCQ	B	1
11	1MCQ	C	1
12	1MCQ	A	1
13	1NAT	C	1

14	1MCQ	B	1
15	1MCQ	A	1
16	1MCQ	B	1
17	1MCQ	B	1
18	1NAT	A	1
19	1NAT	C	1
20	1MCQ	A	1
21	1MCQ	A	1
22	1MCQ	A	1
23	1MCQ	B	1
24	1MCQ	A	1
25	1MCQ	B	1
26	1MCQ	A	1
27	1NAT	A	1
28	1MCQ	C	1

29	1MCQ	B	1
30	1MCQ	A	1
31	1MCQ	D	1
32	2NAT	A	2
33	2MCQ	B	2
34	2MCQ	C	2
35	2MSQ	C	2
36	2MCQ	C	2
37	2MCQ	C	2
38	2MCQ	A	2
39	2NAT	A	2
40	2MCQ	A	2
41	2NAT	A	2
42	2MCQ	B	2
43	2NAT	B	2

44	2MCQ	A	2
45	2MCQ	B	2
46	2MCQ	A	2
47	2MSQ	A	2
48	2MCQ	A	2
49	2MCQ	A	2
50	2MCQ	B	2
51	2MCQ	A	2
52	2MSQ	A	2
53	2MCQ	A	2
54	2NAT	C	2
55	2MCQ	A	2
56	2MCQ	B	2
57	2NAT	A	2
58	2NAT	B	2

59	2MCQ	B	2
60	2MCQ	A	2
61	2NAT	A	2
62	3MCQ	B	3
63	2MCQ	A	2
64	2MCQ	A	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Verbal | Difficulty: easy

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: easy

Q5.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: moderate

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Electrical Machines | Topic: Induction Motor | Difficulty: moderate

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Electrical Machines | Topic: Alternators | Difficulty: moderate

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Electrical Machines | Topic: Induction Motor | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Electrical Machines | Topic: Transformers | Difficulty: easy

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Electrical Machines | Topic: DC Machines | Difficulty: easy

Q11.

Answer: C

Solution:

Correct Answer: C

Subject: Power Systems | Topic: Per Unit | Difficulty: moderate

Q12.

Answer: A

Solution:

Correct Answer: A

Subject: Power Systems | Topic: Generators | Difficulty: easy

Q13.

Answer: C

Solution:

Correct Answer: C

Subject: Power Systems | Topic: Protection | Difficulty: moderate

Q14.

Answer: B

Solution:

Correct Answer: B

Subject: Power Systems | Topic: Transmission | Difficulty: moderate

Q15.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Time Response | Difficulty: moderate

Q16.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Root Locus | Difficulty: moderate

Q17.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Controllers | Difficulty: easy

Q18.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Frequency Response | Difficulty: moderate

Q19.

Answer: C

Solution:

Correct Answer: C

Subject: Power Electronics | Topic: Rectifiers | Difficulty: moderate

Q20.

Answer: A

Solution:

Correct Answer: A

Subject: Power Electronics | Topic: Choppers | Difficulty: moderate

Q21.

Answer: A

Solution:

Correct Answer: A

Subject: Power Electronics | Topic: Converters | Difficulty: difficult

Q22.

Answer: A

Solution:

Correct Answer: A

Subject: Power Electronics | Topic: Inverters | Difficulty: easy

Q23.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Electronics | Topic: Feedback | Difficulty: easy

Q24.

Answer: A

Solution:

Correct Answer: A

Subject: Analog Electronics | Topic: Op-Amps | Difficulty: moderate

Q25.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Electronics | Topic: Oscillators | Difficulty: easy

Q26.

Answer: A

Solution:

Correct Answer: A

Subject: Analog Electronics | Topic: Transistors | Difficulty: easy

Q27.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Electronics | Topic: PLD | Difficulty: moderate

Q28.

Answer: C

Solution:

Correct Answer: C

Subject: Digital Electronics | Topic: Logic Gates | Difficulty: moderate

Q29.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Electronics | Topic: Flip-Flops | Difficulty: moderate

Q30.

Answer: A

Solution:

Correct Answer: A

Subject: Signals and Systems | Topic: Sampling | Difficulty: moderate

Q31.

Answer: D

Solution:

Correct Answer: D

Subject: Signals and Systems | Topic: LTI Systems | Difficulty: moderate

Q32.

Answer: A

Solution:

Correct Answer: A

Subject: Signals and Systems | Topic: Fourier | Difficulty: difficult

Q33.

Answer: B

Solution:

Correct Answer: B

Subject: EMFT | Topic: Maxwell | Difficulty: moderate

Q34.

Answer: C

Solution:

Correct Answer: C

Subject: EMFT | Topic: Transmission | Difficulty: moderate

Q35.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: A

Solution:

Short line: only series Z. $Z = (0.1+0.5)*100 = 60$ ohm. $A=1$, $B=60$, $C=0$, $D=1$.

Subject: Power Systems | Topic: Power Systems | Difficulty: easy

Q39.

Answer: A

Solution:

IGBT: BJT (high current) + MOSFET (easy drive). High input impedance, low saturation voltage.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: moderate

Q40.

Answer: A

Solution:

$V_{out} = 10*5/(5+5) = 5V$. $I = 5/5 = 1A$.

Subject: Network Theory | Topic: Network Theory | Difficulty: difficult

Q41.

Answer: A

Solution:

Gain margin: at phase crossover frequency (phase = -180 deg), $1/|GH(j\omega)|$ in dB.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q42.

Answer: B

Solution:

$N_{sync} = 120*50/4 = 1500$. $s = (1500-1440)/1500 = 60/1500 = 0.04 = 4\%$.

Subject: Power Systems | Topic: Power Systems | Difficulty: moderate

Q43.

Answer: B

Solution:

Ferranti effect: light load/capacitive charging causes $V_s > V_r$. Long lines at no-load.

Subject: Power Systems | Topic: Power Systems | Difficulty: moderate

Q44.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Measurements | Topic: Measurements | Difficulty: moderate

Q45.

Answer: B

Solution:

Current Source Inverter: input is current source (large inductor in series).

Subject: Power Electronics | Topic: Power Electronics | Difficulty: moderate

Q46.

Answer: A

Solution:

$a = 6600/660 = 10$. Primary:secondary = 10:1.

Subject: Electrical Machines | Topic: Electrical Machines | Difficulty: moderate

Q47.

Answer: A

Solution:

Full-wave: reverse voltage across non-conducting diode = V_m .

Subject: Power Electronics | Topic: Power Electronics | Difficulty: moderate

Q48.

Answer: A

Solution:

$h_{11} = (V_1/I_1)$ with output shorted. Units: $V/A = \text{Ohm}$.

Subject: Network Theory | Topic: Network Theory | Difficulty: easy

Q49.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q50.

Answer: B

Solution:

Over-excitation: generator supplies reactive power, operates at leading power factor.

Subject: Electrical Machines | Topic: Electrical Machines | Difficulty: moderate

Q51.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q52.

Answer: A

Solution:

Buck (step-down): $V_o = D \cdot V_{in}$ where D is duty cycle.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: difficult

Q53.

Answer: A

Solution:

$I_{sh} = 220/220 = 1A$. $I_a = 49A$. $E_b = 220 - 49 \cdot 1 = 171V$.

Subject: Electrical Machines | Topic: Electrical Machines | Difficulty: moderate

Q54.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant.

Subject: Power Systems | Topic: Power Systems | Difficulty: moderate

Q55.

Answer: A

Solution:

$a = V_1/V_2 = 6600/660 = 10$. Primary:secondary = 10:1.

Subject: Power Systems | Topic: Power Systems | Difficulty: moderate

Q56.

Answer: B

Solution:

Integral control: eliminates steady-state error for step/ramp inputs. Increases system type by 1.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q57.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q58.

Answer: B

Solution:

Phase margin: at gain crossover ($|GH| = 1$), $1 +$ phase in degrees.

Subject: Control Systems | Topic: Control Systems | Difficulty: difficult

Q59.

Answer: B

Solution:

Root at $s = -1$ (on left half plane). Simple pole on negative real axis: stable (marginally, as $t \rightarrow \infty$).

Subject: Control Systems | Topic: Control Systems | Difficulty: easy

Q60.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Measurements | Topic: Measurements | Difficulty: difficult

Q61.

Answer: A

Solution:

$V_o = \alpha * V_s = 0.6 * 200 = 120V$.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: easy

Q62.

Answer: B

Solution:

Dy connection: primary delta, secondary star. Common for step-down (reduces voltage, provides neutral).

Subject: Electrical Machines | Topic: Electrical Machines | Difficulty: difficult

Q63.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: A

Solution:

Superposition requires linearity (homogeneity + additivity). Does NOT apply to non-linear.

Subject: Network Theory | Topic: Network Theory | Difficulty: difficult

Q65.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 9 questions, 17 marks
Electrical Machines: 9 questions, 14 marks
Power Systems: 9 questions, 14 marks
Power Electronics: 9 questions, 14 marks
Control Systems: 8 questions, 12 marks
Network Theory: 3 questions, 6 marks
General Aptitude: 4 questions, 4 marks
Analog Electronics: 4 questions, 4 marks
Signals and Systems: 3 questions, 4 marks
EMFT: 2 questions, 4 marks
Measurements: 2 questions, 4 marks
Digital Electronics: 3 questions, 3 marks

Difficulty Distribution:

Easy: 16 | Moderate: 41 | Difficult: 8

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.